ESTR1002 Fall 2025

Lab Exercise #2

Due: 17 Sept 2025

Separate your answer for each question in a different .c file. Compress all files into a .zip and submit it to Blackboard.

Question 1

Write a C program to read a length measure in inches (in) (an integer) and convert it to centimeter (cm) (rounded to one decimal place) by using the following formula:

$$cm = 2.54 \times (in)$$

The sample output is shown as follows (the underlined characters are the inputs from the user).

```
10
25.4
```

Question 2

Write a C program to compute the time (*t*) in second of an apple falling onto the ground from a tree height (*H*) in meter. We assume that the gravitational acceleration (*g*) equals to 9.8 m/s^2. You should compute the time (rounded to 2 decimal places) by using the following formula:

$$H = \frac{1}{2} g t^2.$$

Note: You may use the math function “sqrt” by first adding “#include <math.h>” as the first line in your code. The height (*H*) is in the form of the floating-point number.

Input: the height of the tree: *H* in meter.
Output: the time: *t* in second.

The sample output is shown as follows (the underlined characters are the inputs from the user).

```
1.0
0.45
```

Question 3

Write a C program to convert a four-bit binary number to a decimal number. For example, "1011" in binary number system means $1 * 2^3 + 0 * 2^2 + 1 * 2^1 + 1 * 2^0$, which equals to 11, in decimal form.

To do so, you are advised to use the modulo arithmetic, which is an operation that takes the remainder of an integer division for a particular base. It is conveniently represented as x % y in C language. In this formula, it corresponds to extracting the remainder of x when being divided by y, for example 10 % 4 = 2. Modulo arithmetic has some kinds of magic in it, e.g., try N % 10 for any positive integer N. (Note: you need not use looping._

Input: An integer N (0<=N<=1111) in binary form (i.e., each digit is either zero or one).
Output: Decimal form of N.

The sample output is shown as follows (the underlined characters are the inputs from the user).

Example 1:

```
1111
15
```

Example 2:

```
0011
3
```

Question 4

Write a C program to convert a decimal number (ranged from 0 to 15) and print out a four-digit binary number. This question is the inverse of Question 3. (Note: again, you need not use looping.)

Input: An integer N (0<=N<=15) in decimal form.
Output: Binary form of N.

The sample output is shown as follows (the underlined characters are the inputs from the user).

Example 1:

```
15
1111
```

Example 2:

```
3
0011
```